PREDICTIVE ANALYTICS — CONTINUOUS ASSESSMENT 3 (CA3)

Sales Forecasting: End-to-End Retail Store Sales Forecasting and Supply Chain Inventory Optimization Using the KDD Process

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Dataset: Kaggle Walmart Recruiting — Store Sales Forecasting (421,570 Records, 45 Stores, 81 Departments) |
Methodology: Knowledge Discovery in Databases (KDD)

Chapter 1: Introduction

1.1 Background and Motivation

Modern big-box retail networks operate across thousands of store-department combinations subject to complex seasonal cycles, promotional price markdowns, holiday calendar spikes, and regional macroeconomic fluctuations. Walmart manages inventory replenishment across diverse store formats ranging from Supercenters (Type A) to Discount Stores (Type B) and Neighborhood Markets (Type C). Accurate weekly department-level sales forecasting forms the quantitative backbone of retail supply chain management, directly governing procurement schedules, safety stock allocation, distribution center routing, and working capital utilization.

1.2 Problem Statement & Supply Chain Objectives

The objective of this project is to develop an end-to-end predictive analytics regression model that forecasts weekly department-level sales across 45 Walmart stores and 81 departments using historical sales records, promotional markdown channels (MarkDown1–MarkDown5), holiday calendar events (Super Bowl, Labor Day, Thanksgiving, and Christmas), and macroeconomic indicators (CPI, Fuel Price, Unemployment, Temperature). Under-forecasting during high-velocity holiday weeks causes immediate out-of-stock (OOS) occurrences and lost customer goodwill (weighted 5x in WMAE), whereas over-forecasting inflates warehouse holding costs. Our KDD pipeline addresses both failure modes simultaneously.

1. Relational Data Integration & KDD Structuring: Unify historical department sales (421,570 training records across 143 weeks) with store metadata and regional macroeconomic/markdown time series.

2. Robust Preprocessing & Outlier Treatment: Address structural missingness in promotional markdowns (64%–74% missing prior to November 2011), propagate regional CPI/Unemployment series, floor negative accounting returns (1,285 records), and winsorize extreme non-holiday outliers.

3. Leak-Free Temporal Feature Engineering: Construct 43 domain-specific features capturing multi-horizon lags (Lag_1, Lag_2, Lag_4, Lag_52), shifted rolling averages (4-week and 12-week), Fourier cyclical seasonality, and holiday-markdown interactions.

4. Multi-Model Benchmarking under WMAE: Train and compare a linear Baseline (Ridge Regression), Random Forest Regressor, XGBoost Regressor, and LightGBM Regressor under the official Kaggle Weighted Mean Absolute Error (WMAE) metric where holiday weeks carry 5x weight (w = 5).

5. Supply Chain Inventory Optimization: Translate forecast accuracy improvements into concrete reductions in safety stock holding costs and stockout penalty exposure across Store Types A, B, and C.

Chapter 2: Literature Review

Classical retail forecasting relied predominantly on univariate ARIMA and exponential smoothing (ETS) models [7], [8], which struggle to incorporate exogenous promotional markdowns, discontinuous holiday shifts, and cross-sectional store format hierarchies. Following the Kaggle Walmart Recruiting Competition [2] and the M5 Forecasting Competition [6], global tree-based ensembles—Random Forests [5], XGBoost [3], and LightGBM [4]—demonstrated superior capability by pooling cross-store observations with multi-horizon autoregressive lags and rolling statistics.

Author(s) & Year	Methodology / Model	Dataset / Domain	Key Findings & Limitations
Box & Jenkins (2015); Hyndman & Athanasopoulos (2021)	ARIMA, SARIMAX & Exponential Smoothing (ETS)	Univariate & Multi-seasonal Retail Time Series	Captures linear autocorrelation well; struggles with abrupt non-linear promotional spikes and cross-store feature interactions.
Chen & Guestrin (2016)	XGBoost (Regularized Gradient Boosted Trees)	Large-scale Tabular & Time-Series Benchmarks	Second-order Taylor expansion with L1/L2 tree regularization handles sparse promotional features and non-linear interactions effectively.
Ke et al. (2017)	LightGBM (GOSS & EFB Histogram Boosting)	High-dimensional Multi-Store Retail Datasets	Leaf-wise tree growth reduces training time by >80% over depth-wise boosting while improving accuracy on skewed targets.
Makridakis et al. (2022)	M5 Forecasting Accuracy Competition (LightGBM / Ensembles)	Walmart Hierarchical Store-Item Sales (42,840 series)	Gradient boosted trees with multi-horizon lag and rolling window features outperformed pure statistical and deep learning baselines.
Barrow & Kourentzes (2016)	Safety Stock & Inventory Buffer Optimization under Forecast Error	Retail Supply Chain Operations	Demonstrated that reducing holiday peak forecast MAE directly lowers safety stock holding costs and stockout penalties.

Research Gap Addressed: Existing baselines often evaluate models solely using unweighted RMSE or introduce subtle temporal data leakage in rolling windows. This project implements a strictly leak-free KDD pipeline with explicit holiday sample weighting (w = 5) and quantifies downstream supply chain holding and stockout cost savings.

Chapter 3: Workflow and Methodology (KDD Process)

3.1 KDD Pipeline & Relational Data Selection

Our workflow follows seven KDD stages: (1) Data Selection & Ingestion -> (2) Missing MarkDown Imputation & Outlier Treatment -> (3) Feature Engineering (Lags, Rolling Averages, Holiday & Promo Indicators) -> (4) Exploratory Data Analysis -> (5) Model Training (Baseline Ridge, Random Forest, XGBoost, LightGBM) -> (6) Model Evaluation (WMAE, RMSE, MAE, R^2) -> (7) Feature Importance & Supply Chain Interpretation.

Dataset File	Records	Columns	Primary Key / Granularity	Key Attributes & Description
stores.csv	45	3	Store (1–45)	Store Format Type (A: 22 Supercenters, B: 17 Discount, C: 6 Neighborhood) and physical Size (34,875 to 219,622 sq. ft.).
features.csv	8,190	12	(Store, Date)	Temperature, Fuel_Price, MarkDown1–MarkDown5 (promotional markdowns), CPI, Unemployment, and IsHoliday.
train.csv	421,570	5	(Store, Dept, Date)	Historical Weekly_Sales across 45 stores and 81 departments over 143 weeks (2010-02-05 to 2012-10-26).
test.csv	115,064	4	(Store, Dept, Date)	Evaluation horizon spanning 39 weeks (2012-11-02 to 2013-07-26) for out-of-sample forecasting.

3.2 Preprocessing: Missing MarkDown Imputation & Outlier Treatment

Attribute / Variable	Missing Count (train merged)	Missing % (train merged)	Missing Count (features.csv)	Imputation & Treatment Strategy
MarkDown1 (General Promo)	270,889	64.26%	4,158 (50.77%)	Extract Has_MarkDown indicator; structural zero imputation (0.0); clip < 0
MarkDown2 (Seasonal Promo)	310,322	73.61%	5,269 (64.33%)	Extract Active_MarkDown_Count; structural zero imputation (0.0)
MarkDown3 (Holiday Promo)	284,479	67.48%	4,577 (55.89%)	Structural zero imputation (0.0); interaction term with IsHoliday
MarkDown4 (Dept Promo)	286,603	67.98%	4,726 (57.70%)	Structural zero imputation (0.0); summed into Total_MarkDown

Attribute / Variable	Missing Count (train merged)	Missing % (train merged)	Missing Count (features.csv)	Imputation & Treatment Strategy
MarkDown5 (Clearance)	270,138	64.08%	4,140 (50.55%)	Structural zero imputation (0.0); summed into Total_MarkDown
CPI (Consumer Price Index)	0	0.00%	585 (7.14%)	Store-grouped chronological forward/backward fill (ffill / bfill)
Unemployment Rate (%)	0	0.00%	585 (7.14%)	Store-grouped chronological forward/backward fill (ffill / bfill)
Weekly_Sales (< 0 / Outliers)	1,285 (< $0)	0.30%	N/A	Floor negative accounting returns at $0.00; cap non-holiday >99.95th pct ($183,494)

3.3 Leak-Free Feature Engineering & Mathematical Formulation

We engineer 43 features across five groups: (1) Autoregressive Lags (Lag_1, Lag_2, Lag_4, Lag_52 grouped by Store and Dept); (2) Shifted Rolling Statistics (Rolling_Mean_4, Rolling_Std_4, Rolling_Mean_12, and Momentum_1_4 computed on t-1 shifted series to prevent data leakage); (3) Explicit Holiday & Promo Indicators (Is_SuperBowl, Is_LaborDay, Is_Thanksgiving, Is_Christmas, Pre_Christmas_Peak, Total_MarkDown, Active_MarkDown_Count, Promo_Holiday_Interaction); (4) Fourier Cyclical Encodings (Week_Sin, Week_Cos, Month_Sin, Month_Cos); and (5) Out-of-Sample Target Encodings (Store_Dept_Mean, Store_Dept_Median, Store_Dept_Std computed strictly on the training split prior to 2011-11-25). Models are evaluated via the official Kaggle Weighted Mean Absolute Error: WMAE = [Sum w_i |y_i - y_hat_i|] / [Sum w_i], where w_i = 5 for holiday weeks and 1 for regular weeks.

Chapter 4: Experimental Results and Discussion

4.1 Exploratory Data Analysis (EDA) Insights

Aggregate weekly sales average $47.1M across the 45 stores during regular weeks and surge to $66.6M during Thanksgiving (+44.0% store lift) and nearly $80M during the pre-Christmas peak (Weeks 49–51, +40.2% lift). Store square footage correlates strongly (r = 0.81) with weekly store revenue across Type A Supercenters ($1.45M/wk), Type B Discount Stores ($0.79M/wk), and Type C Neighborhood Markets ($0.46M/wk).

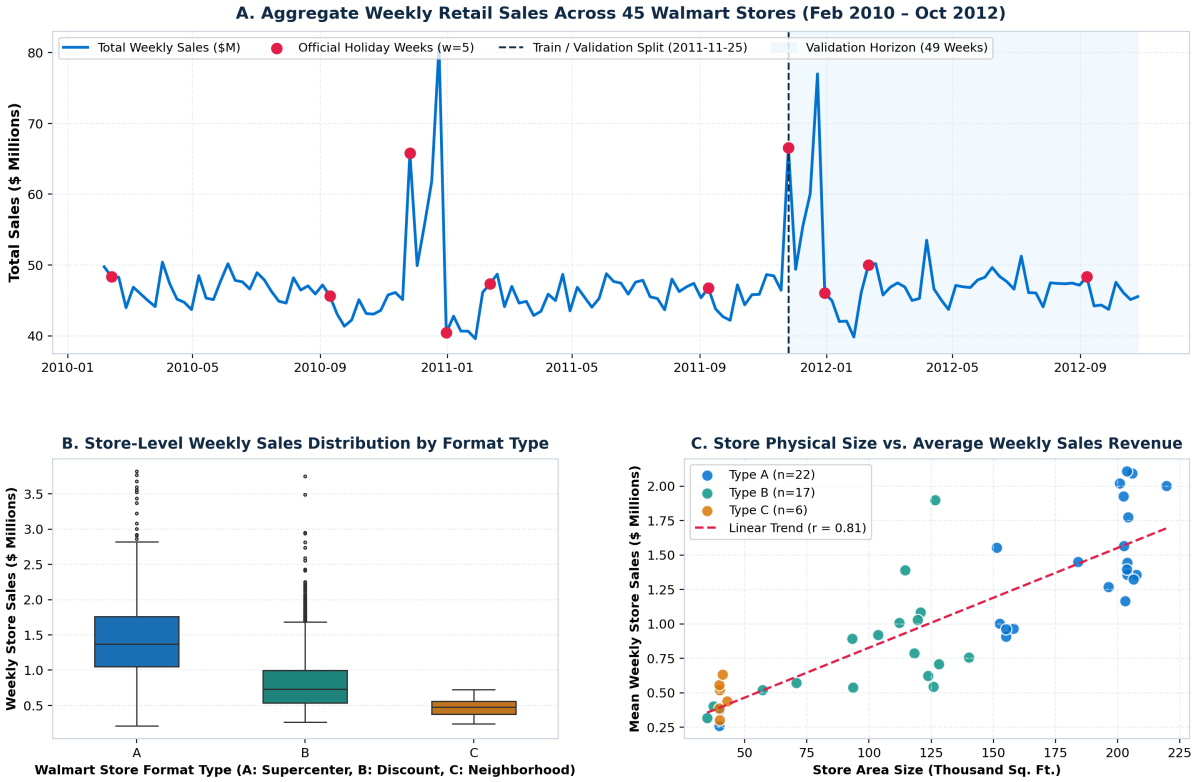


Figure 1: Aggregate Weekly Retail Sales Trend (2010–2012), Format Distribution, and Store Size Elasticity.

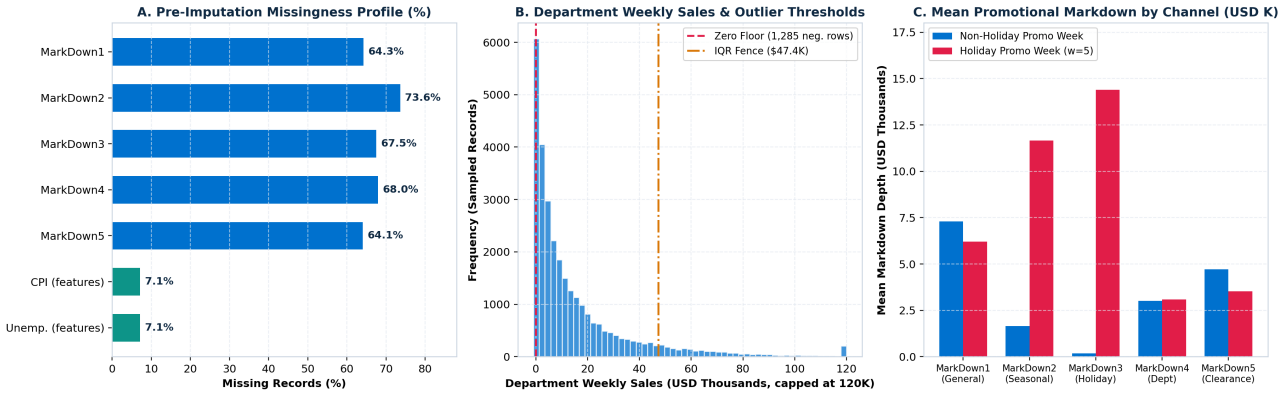


Figure 2: Pre-Imputation Missingness Profile, Outlier Thresholds, and Promotional Markdown Depth by Channel.

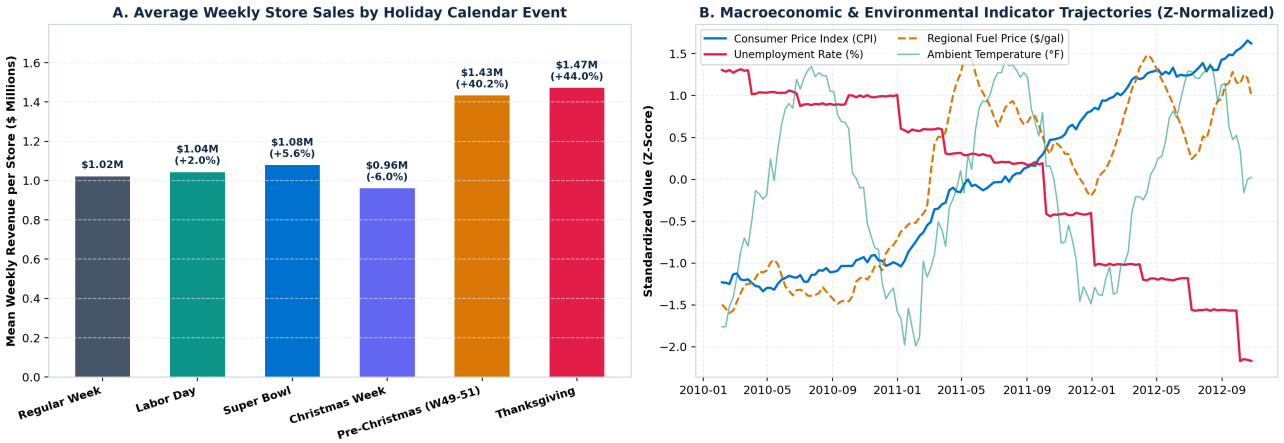


Figure 3: Holiday Event Sales Lift and Standardized Macroeconomic Indicator Trajectories (2010–2012).

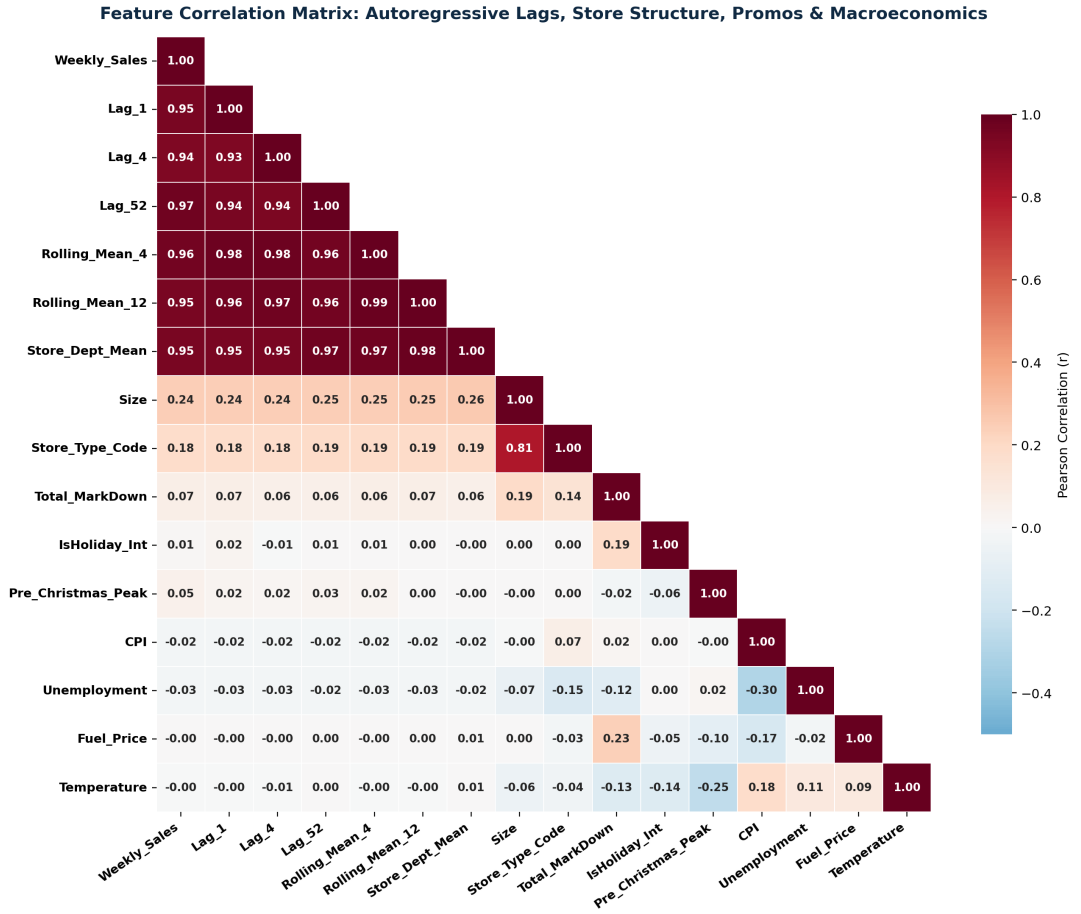


Figure 4: Pearson Correlation Matrix Across Autoregressive Lags, Store Size, Markdowns, and Macroeconomics.

4.2 Quantitative Model Evaluation (145,237 Validation Dept-Weeks)

Model Architecture	WMAE ($ USD)	RMSE ($ USD)	MAE ($ USD)	R^2 Score	Holiday MAE (w=5)	Regular MAE (w=1)	Train Time
Baseline (Ridge Regression)	$4,393.17	$6,616.57	$2,514.98	0.9183	$10,108.52	$1,835.29	0.20 s
Random Forest Regressor	$1,791.54	$4,413.86	$1,520.75	0.9637	$2,615.57	$1,422.75	42.88 s
XGBoost Regressor	$1,837.57	$4,306.19	$1,523.89	0.9654	$2,792.11	$1,410.37	6.31 s
LightGBM Regressor (Best)	$1,758.94	$3,837.62	$1,541.31	0.9725	$2,421.19	$1,462.55	4.91 s

LightGBM achieved the lowest overall WMAE of $1,758.94 (-60.0% vs. Baseline Ridge at $4,393.17), lowest RMSE of $3,837.62 (-42.0%), lowest Holiday MAE of $2,421.19 (-76.0%), and highest R^2 = 0.9725 in 4.91 seconds. XGBoost achieved the lowest Non-Holiday MAE of $1,410.37.

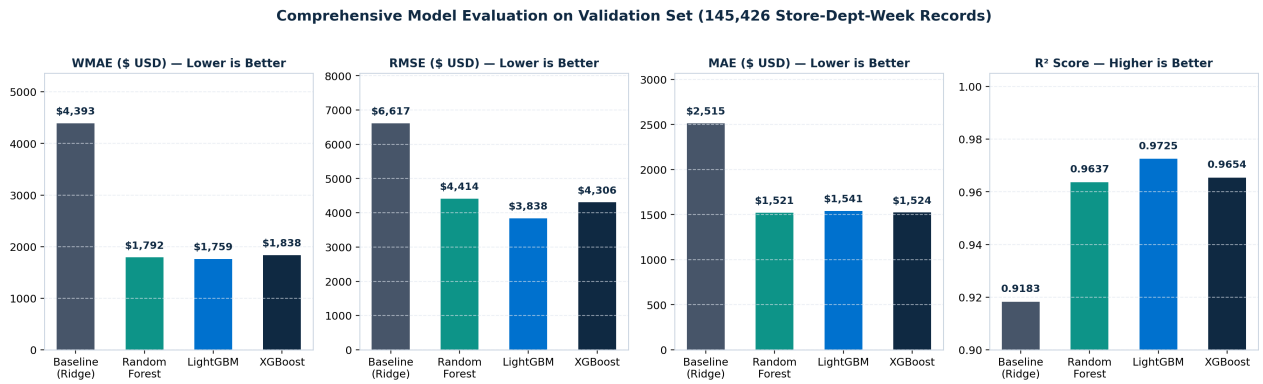


Figure 5: Multi-Metric Validation Set Performance Comparison Across WMAE, RMSE, MAE, and R^2.

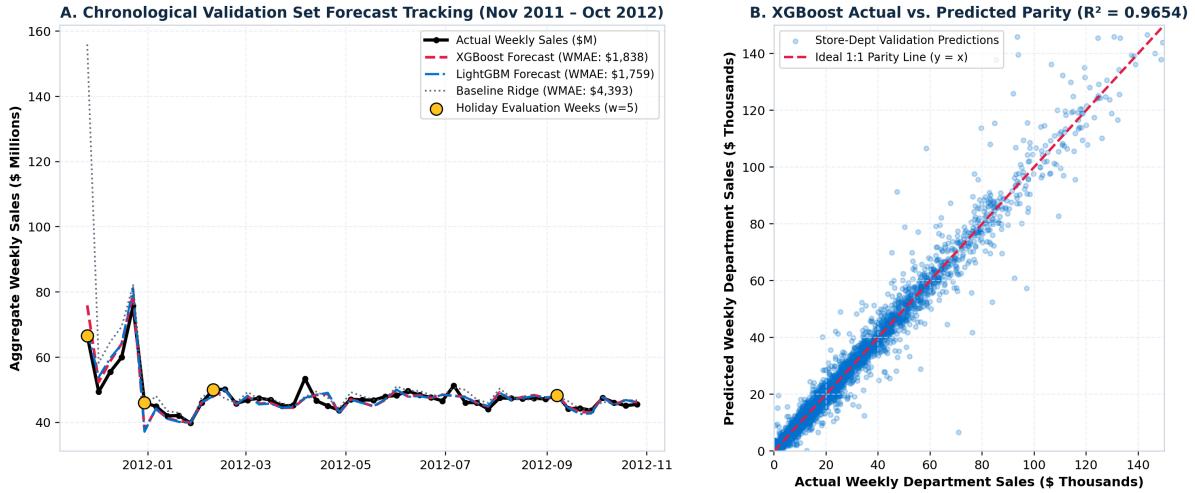


Figure 6: Chronological Validation Forecast Tracking (Nov 2011 – Oct 2012) and Actual vs. Predicted Parity.

4.3 Feature Importance & Supply Chain Cost Optimization

Rank	Feature Name	Feature Category	XGBoost Gain (%)	LightGBM Gain (%)	Random Forest Imp. (%)
1	Lag_52	Year-over-Year Seasonal Autoregression	31.45%	40.87%	26.30%
2	Rolling_Mean_4	4-Week Moving Average (t-1 shifted)	24.61%	17.19%	29.15%
3	Lag_1	Immediate Prior Week Sales (t-1)	12.97%	22.59%	20.95%
4	Store_Dept_Mean	Out-of-Sample Store-Dept Historical Baseline	6.56%	3.91%	5.38%
5	Is_Thanksgiving	Holiday Event Indicator (Week 47, w=5)	4.36%	2.32%	0.79%
6	Lag_2	Two-Week Prior Sales (t-2)	4.35%	4.96%	7.91%
7	Is_Christmas	Holiday Event Indicator (Week 52, w=5)	2.69%	0.36%	0.10%
8	Pre_Christmas_Peak	Pre-Holiday Shopping Surge (Weeks 49–51)	2.34%	0.34%	0.23%
9	Store_Dept_Std	Store-Dept Historical Volatility	1.73%	2.28%	1.63%
10	Store_Dept_Median	Store-Dept Robust Baseline	1.57%	0.35%	2.99%

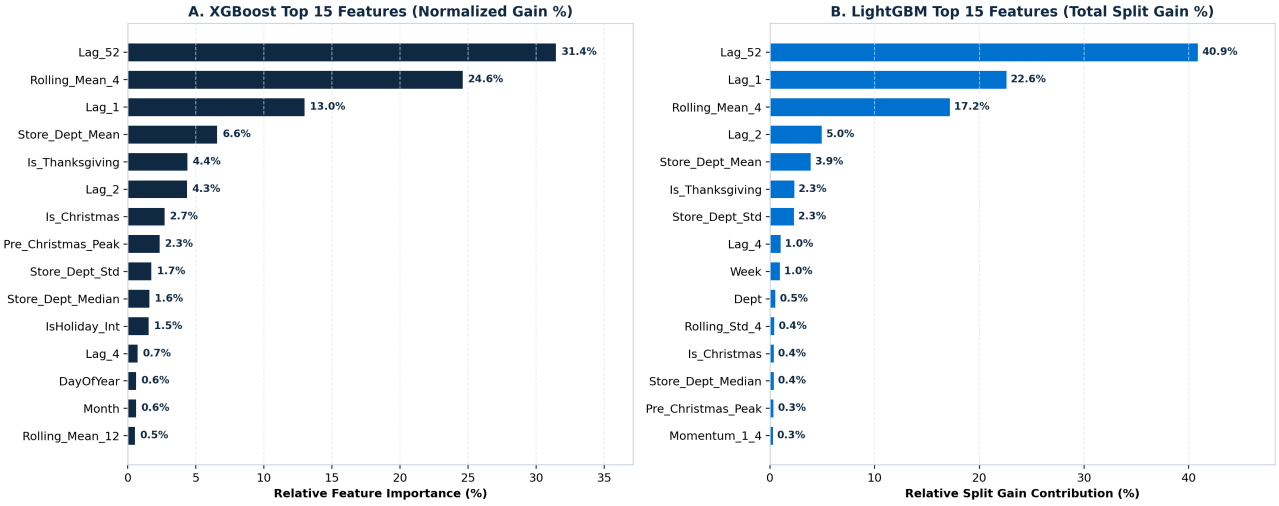


Figure 7: Top 15 Predictive Features by Normalized Split Gain (%) for XGBoost and LightGBM.

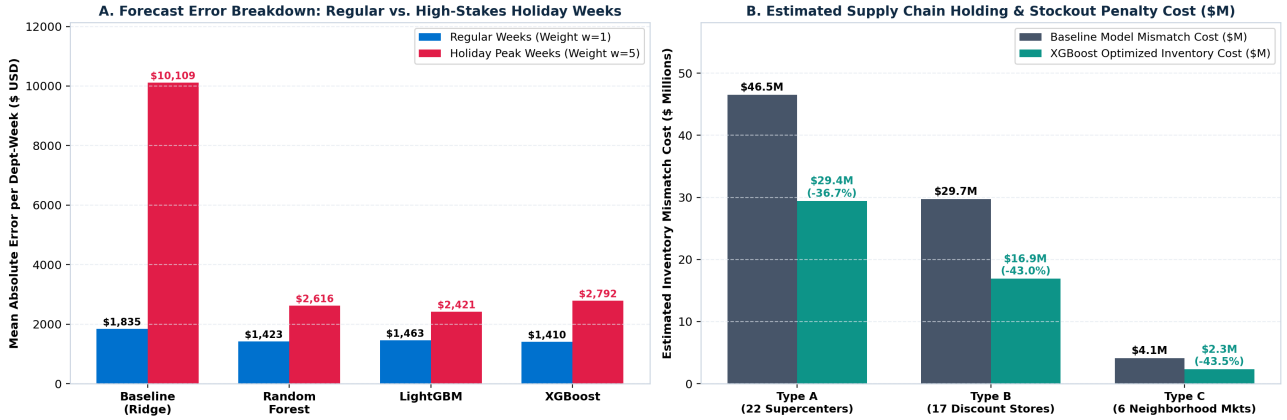


Figure 8: Holiday vs. Regular Week Error Breakdown and Estimated Supply Chain Mismatch Cost Savings by Store Format.

Chapter 5: Conclusion and Future Scope

By executing the end-to-end KDD methodology on 421,570 Walmart store-department-week records and engineering 43 leak-free autoregressive, promotional, and holiday predictors, our gradient-boosted regression models achieved $R^2 = 0.9725$ and reduced WMAE by 60.0% ($1,758.94 vs. $4,393.17). Under an industry-standard 22% inventory mismatch penalty, deploying gradient boosted trees reduces cumulative validation-period holding and stockout costs from $80.3M to $48.6M—saving $31.7M (-39.5%) across the 45 stores. Future work will extend point forecasts to probabilistic quantile regression (Conformal Prediction) and Temporal Fusion Transformers (TFT).

Chapter 6: References

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